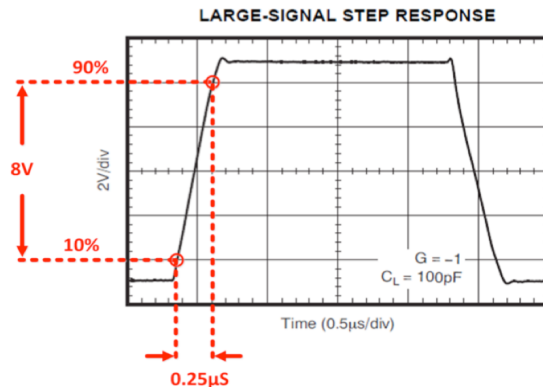


Slew Rate Defined



$$\text{Slew_Rate} = \frac{\Delta V_{\text{out}}}{\Delta \text{Time}} = \frac{(V_{\text{out}90\%} - V_{\text{out}10\%})}{(t_{90\%} - t_{10\%})} = \frac{(9V - 1V)}{(0.625\mu s - 0.375\mu s)} = 29 \frac{V}{\mu s}$$

49

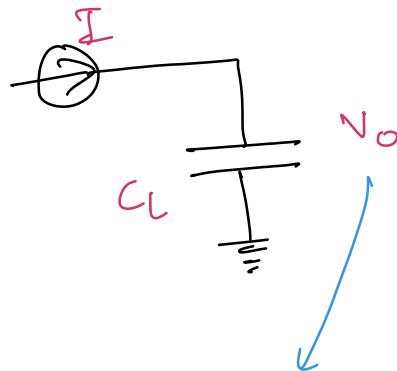
→ So, the Transient analysis can be indeed performed with Laplace Transform but, it neglects some important aspects of our circuit. In particular

the Transfer function is typically calculated by resorting to small signal Analysis.

→ So we Linearize the circuit and estimate the transfer function. However the SSC.A assumes small variations near operating point. In some cases the variations are not negligible.

→ Whenever, we want to charge or discharge

at given capacitances, especially load capacitance we have to supply large current to achieve sufficiently Rapid variation of the Voltage which is applied to the Capacitance.

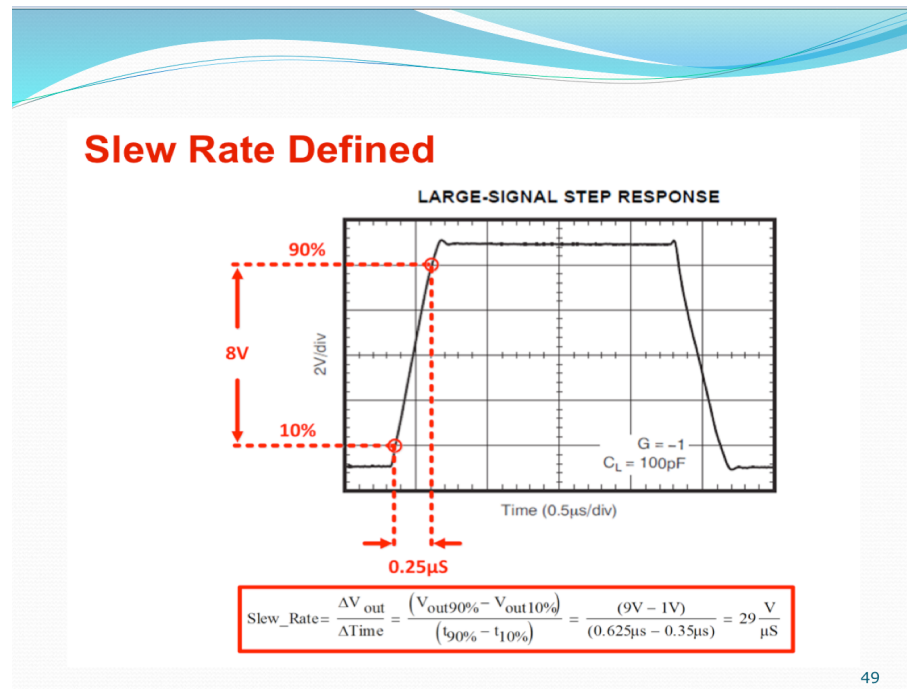


$$q = CV$$

$$\frac{dq}{dt} = C \frac{dV}{dt}$$

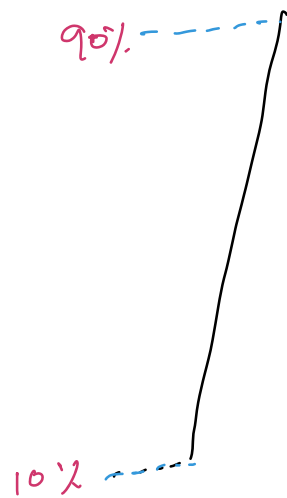
$$\Rightarrow \boxed{i = C \frac{dV}{dt}}$$

The voltage in many practical applications is the ϕ voltage.



^{1>} The
This out Response of a circuit in which we
want the o/p value to vary from a low
value 0V to upto 8V and More.

In This case the slope of the Transition
is very Rapid -



The Transition is very steep

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$$\therefore i_o = C \frac{dv}{dt}$$

If we have High derivative
of the Voltage

i.e Voltage is Rapidly
Changing

i.e current should also
be large

→ Q How big can this current be?
will be a problem.

Because our circuit can't arbitrarily
High currents.

So we have to Investigate this ^{pheno}menon.

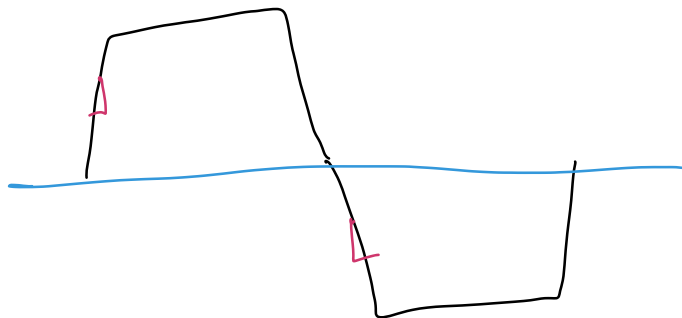
So we basically study the "SLEW RATE" of
the circuit.

$$\text{Slew_Rate} = \frac{\Delta V_{\text{out}}}{\Delta \text{Time}} = \frac{(V_{\text{out}90\%} - V_{\text{out}10\%})}{(t_{90\%} - t_{10\%})} = \frac{(9V - 1V)}{(0.625\mu s - 0.35\mu s)} = 29 \frac{V}{\mu s}$$

This quantity is Intrinsically limited in our
Circuit.

→ Another possible definition of "Slew Rate" is
i.e Maximum of the Derivative of the o/p
voltage w.r.to Time

Typically we are interested in +ve steps &
-ve steps



The distinction b/w +ve & -ve slew rate would be very important when we have a circuit which is not perfectly symmetric.

→ Let's find the slew rate for the topologies we have learned so far.

Slew Rate of the Differential Amplifier

Slew Rate (SR) = Maximum output-voltage rate (either positive or negative)

It is caused by, $i_{OUT} = C_L \frac{dv_{OUT}}{dt}$. When i_{OUT} is a constant, the rate is a constant. Consider the following current-mirror load, differential amplifiers:

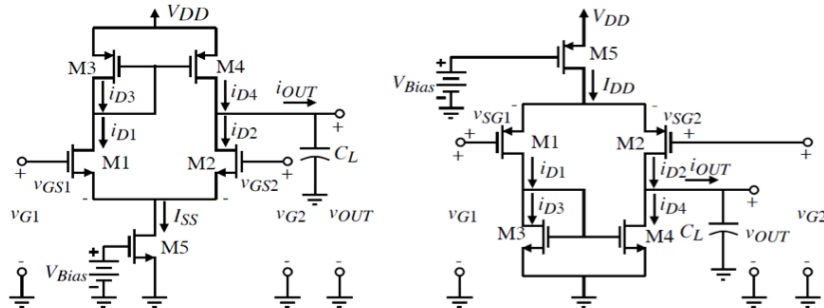


Fig. 5.2-11B

Note that slew rate can only occur when the differential input signal is large enough to cause I_{SS} (I_{DD}) to flow through only one of the differential input transistors.

$$SR = \frac{I_{SS}}{C_L} = \frac{I_{DD}}{C_L} \Rightarrow \text{If } C_L = 5\text{pF and } I_{SS} = 10\mu\text{A, the slew rate is } SR = 2\text{V}/\mu\text{s.}$$

(For the BJT differential amplifier slewing occurs at $\pm 100\text{mV}$ whereas for the MOSFET differential amplifier it can be $\pm 2\text{V}$ or more.)

The Slew rate = $\text{Max} \left(\frac{dv_{out}}{dt} \right)$

$$i_{out} = C_L \frac{dv_{out}}{dt}$$

we can track $\frac{dV_{out}}{dt}$ by tracking the output current i_{out} whether it will achieve the SlewRate specifications.

∴ we are looking at Large signal concept
we can not apply the machinery we have studied for small signal Analysis for calculating SlewRate. SSA will help in studying the Frequency Analysis but SSA is not suitable for SlewRate calculations.

→ The problem is ∵ There is linear relationship between i_{out} & $\frac{dV_{out}}{dt}$

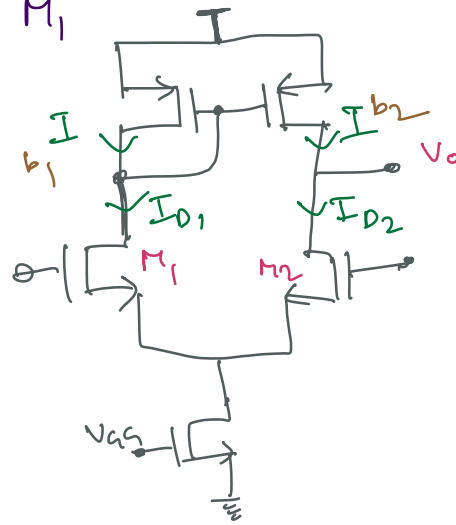
To determine $\max\left(\frac{dV_{out}}{dt}\right)$ we must determine
Maximum i_{out} → SlewRate

→ The maximum current is supplied when we

apply a step Function to one of the input of the Differential pair

*** In a Differential pair is perfectly balanced if same voltage is applied to the positive and negative terminals

Then the current that flows in this device M_1 is the same current that flows in M_2 .

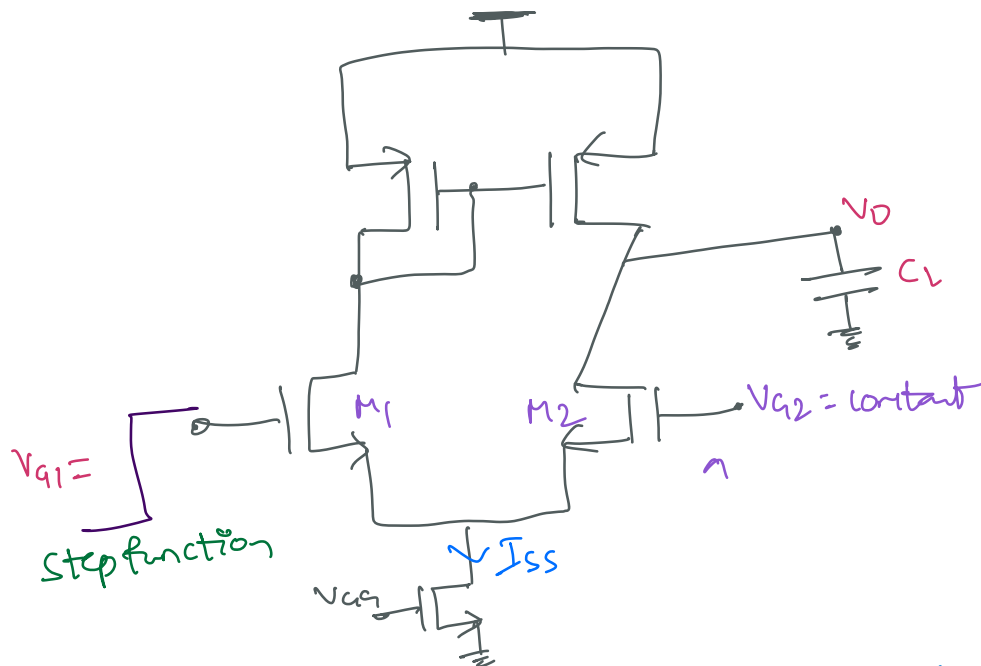


→ Since we have current mirror loading (I) the current in both branches (b_1, b_2) are same and are equal to the current flowing through M_1 and M_2 respectively.

∴ The current flowing into the output is Zero.

→ The situation is Radically different when we apply a step Function to one of the

input of Differential Pair and keep the other input constant.



→ This means that the voltage applied to two branches is different.

With this M_1 becomes more conductive in comparison to M_2 because $V_{G1} > V_{G2}$

So, this implies that I_{SS} current which was earlier split into two equal parts now favours the flow towards Branch 1

→ Basically I_{SS} tends to flow into first branch of the Diff. Pair.

which implies that no current flows through M_2 as a consequence of that

Basically $I_{D2} = 0$, $I_{D1} = I_{SS}$

→ let's see what happens in the upper part of the circuit where we have a Current Mirror. in which M_3 & M_4 are Transistor which are still driven by same V_{SG} .

This Current Mirror does its job without any interruption. $\therefore$ The current $I_{D1} = I_{SS}$

which flows in Branch 1 of the current mirror should also flow into the same branch of the Current Mirror.

$$\text{i.e. } I_{b1} = I_{D1} = I_{SS}$$

$$\Rightarrow I_{b2} = I_{SS} \text{ \& } I_{D2} = 0$$

$\therefore$ all current I_{b2} flows into the output as I_{out}



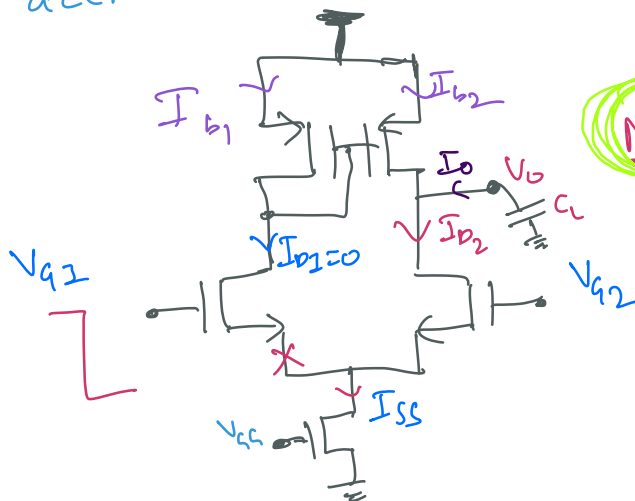
$$\therefore I_{out} = I_{b2} = I_{D1} = I_{SS}$$

$$\frac{I_{out}}{C_L} = \frac{I_{SS}}{C_L} = \left(\frac{dV_{out}}{dt} \right)_{\text{slew rate}}$$

Actually all this was related to the step function.

Q What happens in the negative step function?

If we apply $-ve$ step to the M_1 and M_2 constant now the conductivity of M_1 decreases and the M_2 increases. So I_{SS} increases. M_2



Note

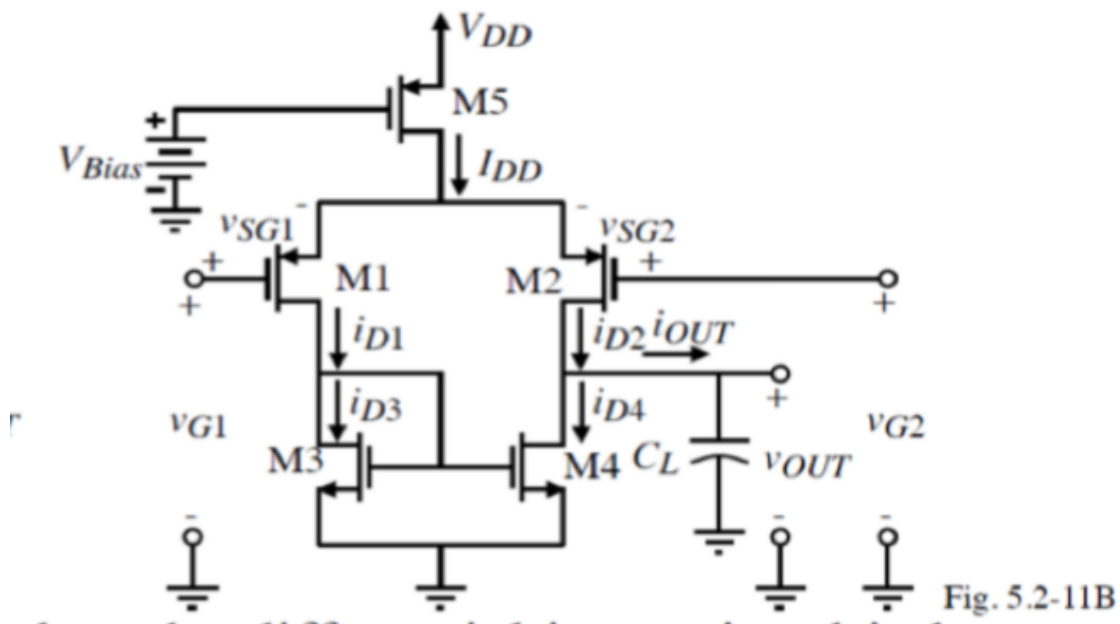
→ The input of the current mirror sees no current and which implies no current at the output.

$$I_{b1} = I_{b2} = 0$$

→ The $I_{SS} = I_{D2} = I_0$ removes the charge accumulated on C_L $\therefore V_O$ decreases

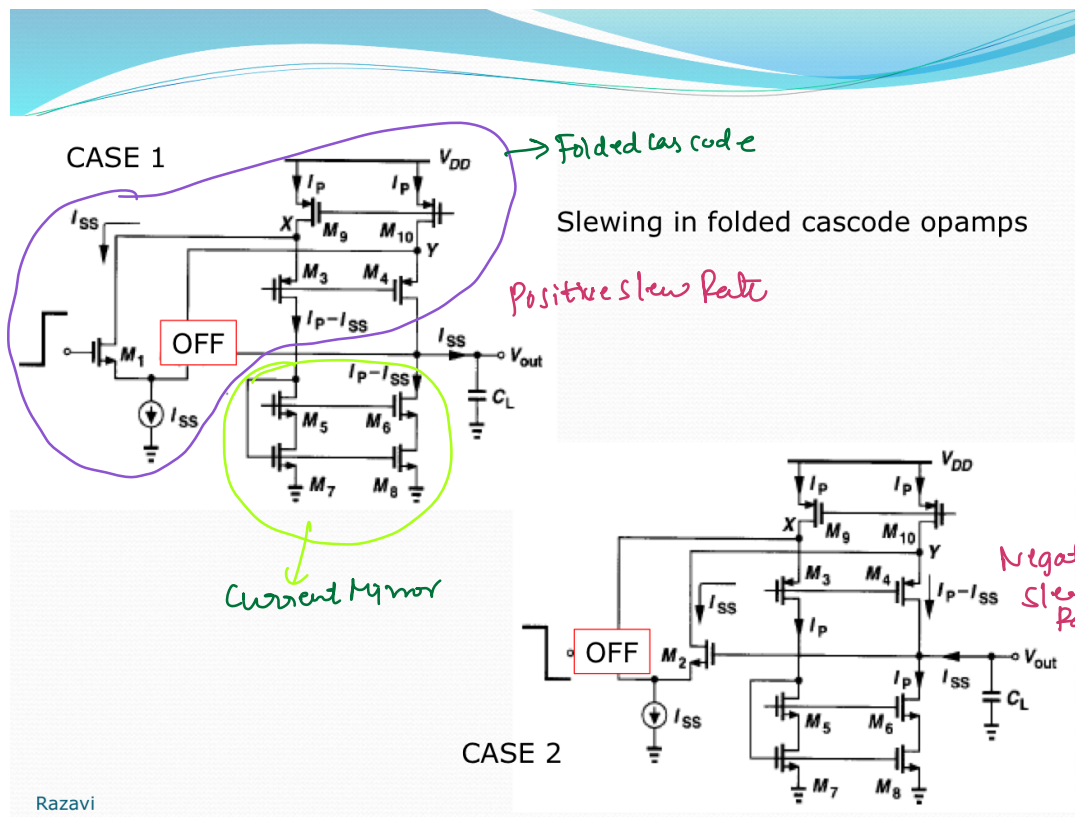
$$\frac{I_{SS}}{C_L} = -\frac{dV_{out}}{dt}$$

Note:- Analysis is similar for pmos Diff-pair



(For the BJT differential amplifier slewing occurs at $\pm 100\text{mV}$ whereas for the MOSFET differential amplifier it can be $\pm 2\text{V}$ or more.)

Slew Rate in Folded cascode op-Amps



Slew Rate
The analysis is similar to what we have done above.

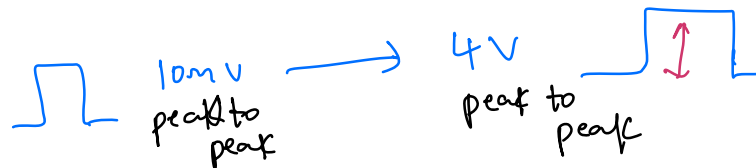
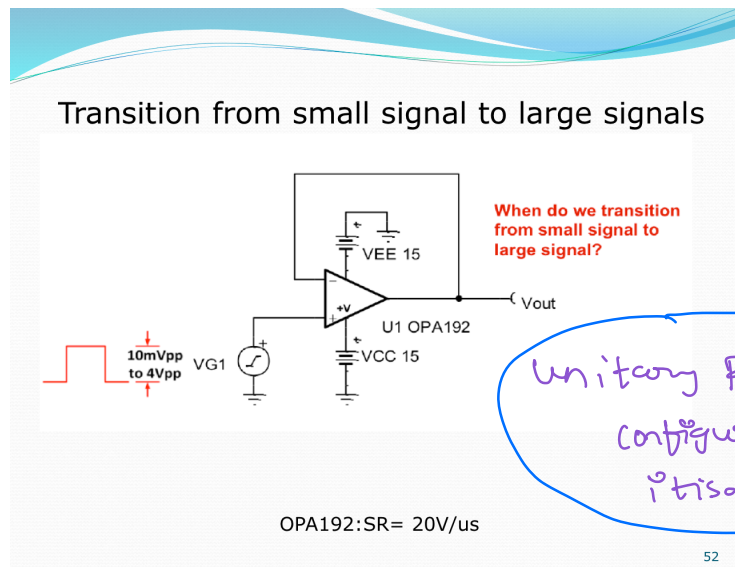
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Numerical example

* To check whether this slew rate limitation somehow changes the op voltage and expectedly wrt conventional transfer function or not.

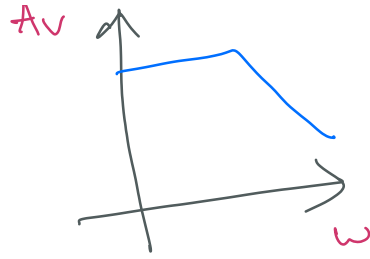
* we have wide Range of situations where Slew Rate High Intervene so we must be careful about it.

ex:-



$\rightarrow$ And we are interested in Transient function of this circuit when we have Unity Feed back configuration. Basically Voltage Buffer

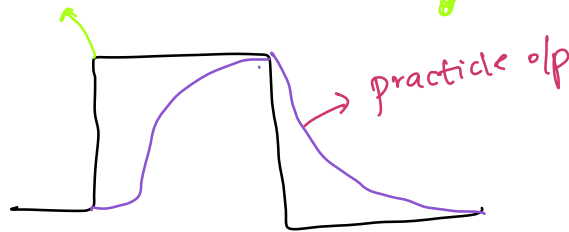
$\rightarrow$ let ^{suppose} the Transfer function of the above device modelled by a single pole.



∴ In principle the device should duplicate i/p to o/p.

This would be true if transfer function of the device is Ideal.

Ideal o/p



→ We want to Investigate how these trends are related to SlewRate Term.

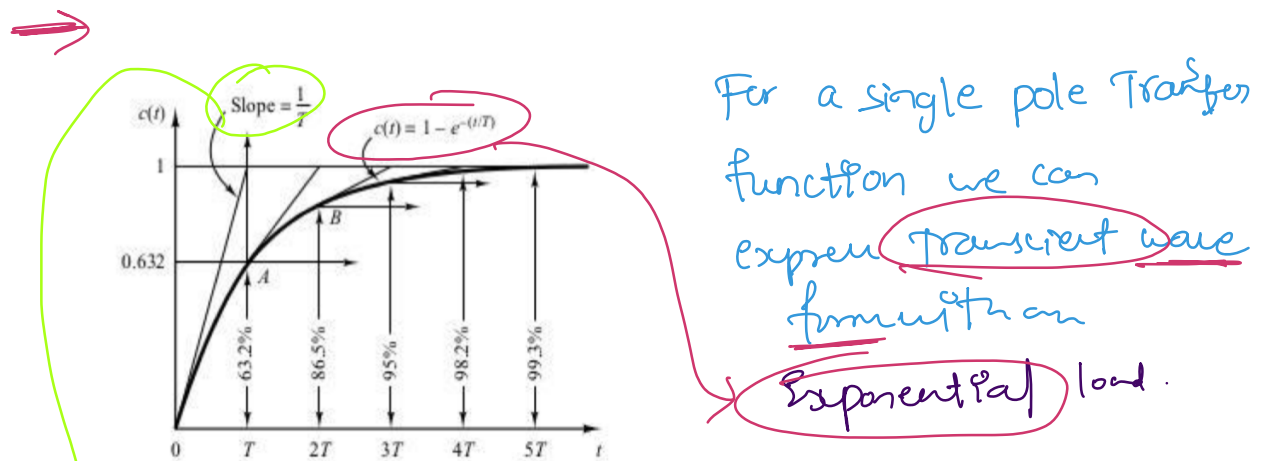
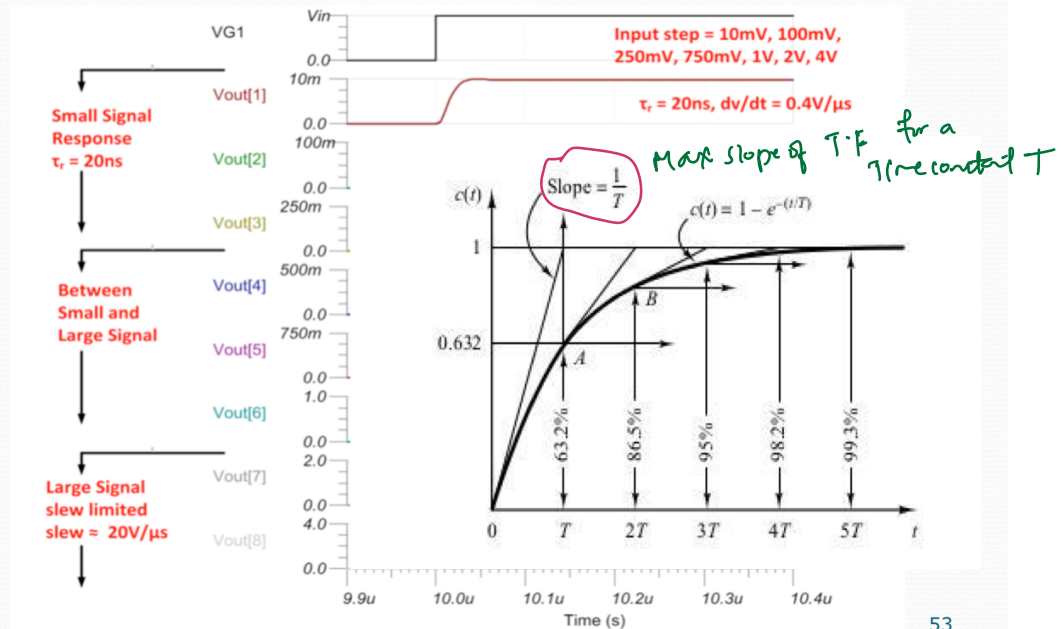
$$\text{Here SlewRate} = 20\text{V}/\mu\text{s}$$

$$\text{OPA192:SR} = 20\text{V}/\mu\text{s} = \frac{20\text{MV}}{\text{Sec}}$$

Q What does This mean in practice?

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Small signal vs. Large signal



This is the typical trend of single pole T.F

Maximum derivative of T.F is given by the

slope $= \frac{1}{T}$ which is related to Time

Constant (T)

The slope becomes smaller if we apply smaller step functions. Slope becomes higher if the step function is higher than ①

② What happens when the amplitude of step function is of different values?

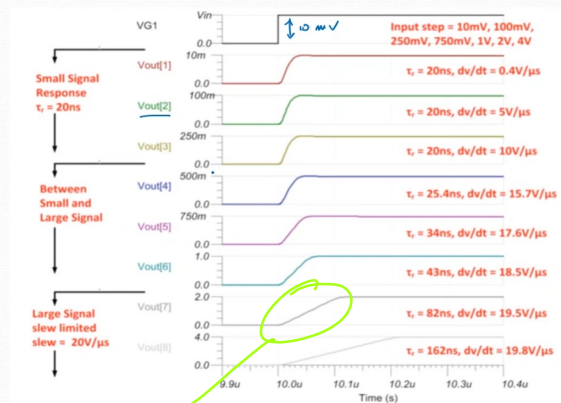


But if we change the amplitude to 100mV

250mV, 750mV, 1V, 2V, 4V

What happens?

Small signal vs. Large signal



Here circuit is same
The only thing that changes is the o/p.

Here we have piecewise linear response of the circuit.

$\therefore$ Slew Rate is a bounding factor that is the slope of the trend.

So for lower voltages, ^{if} intention is side effects can be neglected.

ultimately $i = C_L \frac{dv_{out}}{dt}$

If this quantity is limited

i.e. $SR = 20V/\mu s$

Then for higher input voltages the

o/p to reach its peak will take more time as we have seen above. Because there is a current limitation because of slew rate.

Design of Differential Amplifiers

We will combine all the concepts we have studied till now to implement the Design of Diff. Amplifier - starting from specifications to Determine the Right Sizing of the Transistors -

→ which is the reverse of what we have done till now.

Till now we had transistors of specific size and we found Gain, Slew Rate, ICMR, Frequency Response. Now we do the opposite.



DESIGN OF DIFFERENTIAL AMPLIFIERS

Design of a CMOS Differential Amplifier with a Current Mirror Load

Design Considerations:

Constraints	Specifications
Power supply	Small-signal gain
Technology	Frequency response (C_L)
Temperature	ICMR
	Slew rate (C_L)
	Power dissipation

Relationships

$$A_v = g_{m1} R_{out}$$

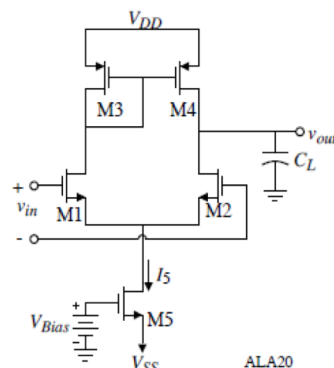
$$\omega_{-3dB} = 1/R_{out} C_L$$

$$V_{IC(max)} = V_{DD} - V_{SG3} + V_{TN1}$$

$$V_{IC(min)} = V_{SS} + V_{DS5(sat)} + V_{GS1} = V_{SS} + V_{DS5(sat)} + V_{GS2}$$

$$SR = I_{SS}/C_L$$

$$P_{diss} = (V_{DD} + |V_{SS}|) \times (\text{All dc currents flowing from } V_{DD} \text{ or to } V_{SS})$$



→ let's review the Diff. Amplifier specification and characteristics and determine Design strategy from this specification.

→ First of all in our circuit There are some constraints Directly posed by Technologies which is available to us.

→ Depending on the technology our company can use, we are forced to set

- Power supply, Intrinsic conductive,

Threshold voltages. These are parameters which are not in our control.

In most cases the Technology contract is given to us in advance.

Constraints

Power supply

Technology

Temperature

→ Specifications which are given to us.
which ought to be performed by the
Designed circuit.

Specifications

Small-signal gain

Frequency response (C_L)

ICMR

Slew rate (C_L)

Power dissipation

→ what have we learned about the
Differential Amplifier-

$$V_{DS1} \Rightarrow V_{in} - V_{OV} - V_T$$

($V_{S1} = V_{OV}$)
gère

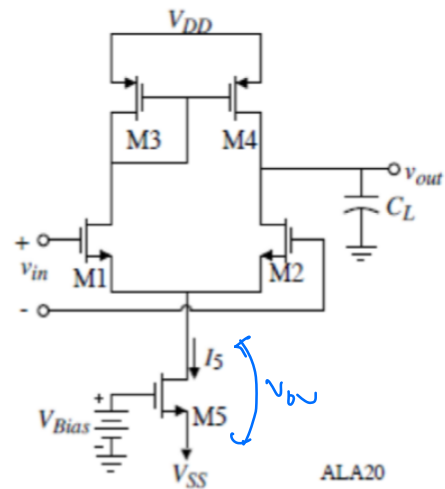
$$\Rightarrow V_{D1} - V_{S1} = V_{in} - V_{OV} - V_T$$

$$\Rightarrow V_{DD} - V_{GS3} - \cancel{V_{OV}} = \cancel{V_{in}} - \cancel{V_{OV}} - V_T$$

$$\Rightarrow V_{in_{Max}} = V_{DD} - V_{GS3} + V_{T_{N1}}$$

2 $V_{in_{min}} = ?$

$$V_{DSS(sat)} =$$



$$V_{GS3} - V_T = V_{OV}$$

$$V_{GS3} - V_T = V_{OV}$$

$$V_{DSS(sat)} = V_{DS} - V_{SS}$$

$$V_{DS(sat)} = V_{S1} = V_{DSS(sat)} + V_{SS}$$

$$V_{GS1} - V_T = V_{OV}$$

$$\Rightarrow V_{G1} - V_{S1} - V_T = V_{OV}$$

..

$$\Rightarrow V_{inmin} - V_{S1} - V_T = V_{GS1} - V_T$$

$$\Rightarrow V_{inmin} - V_{DSS}^{(sat)} - V_{SS} - V_T = V_{GS1} - V_T$$

$$\Rightarrow V_{inmin} = V_{GS1} + V_{DSS}^{(sat)} + V_{SS}$$

* Slew rate we have studied in this lecture

$$= \frac{I_{SS}}{C_L}$$

$$\star \text{ Power Dissipation} = (V_{DD} + |V_{SS}|) \times$$

(All dc currents flowing from V_{DD} (or to V_{SS}))

ee
slew rate & Power dissipation
are intrinsically conflicting.

→ Q If we are given below specifications. How would

You approach the Design of Differential Amplifier. ?

Design constraints & specifications

Circuit description

$$C_L = 5 \text{ pF}$$

$$V_{DD} = -V_{SS} = 2.5 \text{ V}$$

$$SR = 10 \text{ V}/\mu\text{s}$$

$$f_{-3\text{dB}} \geq 100 \text{ kHz}$$

$$DC_{\text{gain}} = 100$$

$$-1.5 \text{ V} \leq V_{in_{CM}} \leq 2 \text{ V}$$

$$\text{Power}_{\text{loss}} \leq 1 \text{ mW}$$

Technology specifications

$$\beta'_n = 110 \mu\text{A}/\text{V}^2$$

$$\beta'_p = 50 \mu\text{A}/\text{V}^2$$

$$V_{TP} = -0.7 \text{ V}$$

$$V_{TN} = 0.7 \text{ V}$$

$$\lambda_n = 0.04 \text{ V}^{-1}$$

$$\lambda_p = 0.05 \text{ V}^{-1}$$

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⇒ we will first proceed with the calculation of
"Slew Rate"

DESIGN OF DIFFERENTIAL AMPLIFIERS

Design of a CMOS Differential Amplifier with a Current Mirror Load

Design Considerations:

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Power supply	Small-signal gain
Technology	Frequency response (C_L)
Temperature	ICMR
	Slew rate (C_L)
	Power dissipation

Relationships

$$A_v = g_{m1} R_{out}$$

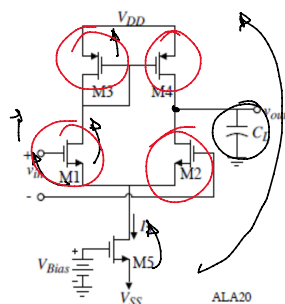
$$\rightarrow \omega_{-3dB} = 1/R_{out} C_L$$

$$\rightarrow V_{IC(max)} = V_{DD} - V_{SG3} + V_{TN1}$$

$$\rightarrow V_{IC(min)} = V_{SS} + V_{DS5(sat)} + V_{GS1} = V_{SS} + V_{DS5(sat)} + V_{GS2}$$

$$\rightarrow SR = I_{SS}/C_L$$

$$\rightarrow P_{diss} = (V_{DD} + |V_{SS}|) \times (\text{All dc currents flowing from } V_{DD} \text{ or to } V_{SS})$$



$$C_L = 5 \text{ pF}$$

$$V_{DD} = -V_{SS} = 2.5 \text{ V}$$

$$SR \gg 10 \text{ V}/\mu\text{s}$$

$$f_{-3dB} \gg 100 \text{ kHz}$$

$$DC_{gain} = 100$$

$$-1.5 \leq V_{in,CM} \leq 2 \text{ V}$$

$$P_{diss} \leq 1 \text{ mW}$$

$$\beta'_n = 110 \mu\text{A}/\text{V}^2$$

$$\beta'_p = 50 \mu\text{A}/\text{V}^2$$

$$V_{TP} = -0.7 \text{ V}$$

$$V_{TN} = 0.7 \text{ V}$$

$$\lambda_n = 0.04 \text{ V}^{-1}$$

$$\lambda_p = 0.05 \text{ V}^{-1}$$

$$\therefore I_5 = I_{SS}$$

$$1) \quad SR = \frac{I_5}{C_L} \Rightarrow I_5 \geq 50 \mu\text{A} = SR \times C_L$$

we now have to find current b/w 50μA, 200μA

$$* \quad P_{diss} \leq 1 \text{ mW} \Rightarrow 5 \text{ V} \cdot I_5 \Rightarrow I_5 \leq 200 \mu\text{A}$$

power dissipation

$$2) \quad \omega = 2\pi f = \frac{1}{R_{out} C_L} \Rightarrow R_{out} \leq 318 \text{ k}\Omega$$

$$R_{outM2} = r_{o2} = \frac{1}{g_{o2}}$$

$$R_{outM4} = r_{o4} = \frac{1}{g_{o4}}$$

intrinsic Resistances

$$R_{out} = R_{outM2} \parallel R_{outM4} \rightarrow \text{parallel b/w } R_{outM4} \text{ \& } R_{outM2}$$

$$g_o = \lambda \cdot I$$

$$R_{out} = \frac{1}{g_{o2} + g_{o4}} = \frac{1}{\lambda_n \cdot I_2 + \lambda_p \cdot I_4}$$

$$I_2 = I_4 = \frac{I_5}{2}$$

perfect balanced condition

$$318 \text{ k}\Omega = R_{out} = \frac{1}{(\lambda_p + \lambda_n) \cdot \frac{I_5}{2}} \Rightarrow I_5 \geq 70 \mu\text{A}$$

now we have got additional constraint for I_5

let

$$\rightarrow I_5 = 100 \mu\text{A}$$

which is in the feasible Region

and it's arbitrary selection and we can change it future.

$$70 \mu\text{A} < I_5 < 200 \mu\text{A}$$

Common Mode input Maximum voltage

$$3) \quad V_{IC,MAX} = V_{DD} - V_{SG3} + V_{TN}$$

$$\rightarrow 0 \text{ V} = 2.5 \text{ V} - V_{SG2} + 0.7 \text{ V}$$

$$\frac{I_5}{2}$$

$$I = \frac{\beta}{2} \cdot V_{ov}^2$$

$$\beta = \beta' \cdot \frac{W}{L}$$

In nominal condition we have equal splitting of current

$$2V = 2.5V - V_{SG3} + 0.7V$$

$$V_{SG3} = 1.2V = V_{ov3} + |V_{TP}| = \sqrt{\frac{2 I_3}{\beta'_p \left(\frac{W}{L}\right)_3}} + |V_{TP}| = 1.2V$$

Because the current mirror should be perfectly balanced

$$\frac{W_3}{L_3} = 8 = \frac{W_4}{L_4}$$

From this expression we found the W/L Ratios of M_3 & M_4 Transistors

$$4) A_V = 100 = g_{m1} R_{out} = \frac{g_{m1}}{g_{o2} + g_{o4}} = \frac{\sqrt{2 I_1 \beta_1}}{(\lambda_n + \lambda_p) \frac{I_5}{2}} = \frac{\sqrt{2 \beta_1 \frac{W_1}{L_1}}}{(\lambda_p + \lambda_n) \sqrt{\frac{I_5}{2}}} = 100$$

$$\frac{W_1}{L_1} = 18.4 = \frac{W_2}{L_2}$$

W/L Ratios of M_1 & M_2

$$5) V_{IN_{CM_{min}}} = V_{SS} + V_{ov5} + V_{ov1} + V_{TN}$$

$$-1.5V = -2.5V + V_{ov5} + \sqrt{\frac{2 I_1}{\beta'_n \frac{W_1}{L_1}}} + 0.7V$$

$$I_1 = \frac{I_5}{2}$$

$$V_{ov5} = 0.078V$$

$$I_5 = 100 \mu A$$

$$\left(\frac{W}{L}\right)_5 = 150.6$$

$$V_{ov}, I_D, \frac{W}{L}$$

$$V_{IN_1} - V_{S_1} = V_{ov1} + V_{TN}$$

$$V_{IN_{CM_{min}}} = V_{S_1} + V_{ov1} + V_{TN}$$

$$V_{S_1} - V_{SS} = V_{osat}$$

$$V_{S_1} = V_{osat} + V_{SS}$$

$$V_{S_1} = V_{ov5} + V_{SS}$$

$$\Rightarrow V_{IN_{CM_{min}}} = V_{SS} + V_{ov5} + V_{ov1} + V_{TN}$$

→ If you observe the w/L ratio of M_5 is larger compared to all the other transistors.

procedure

- First you calculate the Transistor sizings
- Then check with LTSpice.

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